

# Origin of the CEEX–PHOKHARA discrepancy in the pion angular distributions

KLOE large-angle scenario,  $e^+e^- \rightarrow \pi^+\pi^-\gamma$  at NLO,  $F_\pi = 1$ , no vacuum polarisation

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## Abstract

In the KLOE large-angle (KLOE-LA) scenario the NLO predictions of CEEX and PHOKHARA 10.0 for  $e^+e^- \rightarrow \pi^+\pi^-\gamma$  agree in the total cross section and in every charge-even distribution, but disagree in the pion polar angles: the forward–backward asymmetry of the  $\pi^+$  is  $A(\theta^+) = 0.2375$  in CEEX (and 0.2372 in BabaYaga) against 0.2787 in PHOKHARA. We show that the difference is entirely due to two errors in the one-photon ISR×FSR interference at NLO in PHOKHARA (routine `ampL0`, pion branch of the full-NLO mode): a misplaced parenthesis that adds three helicity terms of the tree interference a second time, and a block that keeps the tree interference together with an analytic ISR soft function that the full-NLO mode replaces by the eikonal elsewhere. The second error makes the PHOKHARA asymmetry depend on the unphysical soft-photon cutoff. A point-by-point comparison of the one-photon matrix elements shows that the two errors account for the whole charge-odd difference to  $2.5 \times 10^{-5}$  of the tree. With both corrected, PHOKHARA gives  $A(\theta^+) = 0.23724(10)$  averaged over the cutoff and 0.23725(3) with  $5 \times 10^9$  events, independent of the cutoff and in agreement with CEEX and BabaYaga.

## Contents

|          |                                   |           |
|----------|-----------------------------------|-----------|
| <b>1</b> | <b>Summary of findings</b>        | <b>2</b>  |
| <b>2</b> | <b>Setup</b>                      | <b>2</b>  |
| <b>3</b> | <b>The discrepancy</b>            | <b>3</b>  |
| <b>4</b> | <b>Strategy</b>                   | <b>4</b>  |
| <b>5</b> | <b>Tests</b>                      | <b>5</b>  |
| <b>6</b> | <b>The two errors in PHOKHARA</b> | <b>7</b>  |
| <b>7</b> | <b>Result with the fix</b>        | <b>8</b>  |
| <b>8</b> | <b>Caveats and open points</b>    | <b>10</b> |
| <b>9</b> | <b>Other findings</b>             | <b>11</b> |
| <b>A</b> | <b>Reproduction and files</b>     | <b>11</b> |

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\*Analysis carried out with Claude Code (Anthropic). Repository: [https://github.com/jpaltrinieri/CEEX\\_PHOKHARA\\_ANALYSIS](https://github.com/jpaltrinieri/CEEX_PHOKHARA_ANALYSIS)

## 1 Summary of findings

1. **The disagreement is reproduced** with a fresh PHOKHARA build and fresh CEEX runs (Section 3): totals agree to  $10^{-4}$ , charge-even shapes agree, and only  $\theta^\pm$  and  $\theta_{\text{avg}}$  differ ( $\chi^2/\text{ndf} \approx 10^3$ ). At LO the two codes agree exactly.
2. **The PHOKHARA asymmetry depends on the soft cutoff  $w$** :  $A(\theta^+)$  changes by  $+0.0156$  per decade of  $w$  while  $\sigma$  is constant (Section 5.1).
3. **The whole effect sits in the charge-odd part of the one-photon contribution** (Born + virtual + soft); the hard two-photon emission agrees between the codes (Section 5.3).
4. **Two errors in PHOKHARA explain it point by point** (Sections 5.4 and 6):
  - **H1** – `phokhara_10.0.f:9969-9982`: the factor  $(v_f + v_s)$  multiplies only the last of four helicity terms of the tree interference;
  - **H2** – block B5, the ISR one-loop current times the FSR tree (`phokhara_10.0.f:9998-10007`; `ISRcurrent_no_mass_terms.f:134,210`): an extra  $\text{Re}[F^*I](1 + S_{\text{ISR}}(w))$ .

Subtracting both leaves a per-point residual of  $2.5 \times 10^{-5}$  of the tree.

5. **With both corrected**,  $A(\theta^+)$  no longer depends on  $w$  and agrees with CEEX and BabaYaga, in total and separately in the one- and two-photon parts (Section 7). Totals and charge-even distributions are unchanged.

## 2 Setup

**Scenario.** KLOE-LA (Strong2020 KLOE-I),  $\sqrt{s} = 1.02$  GeV:  $50^\circ \leq \theta_\pm \leq 130^\circ$ ;  $|p_\pm^z| > 90$  MeV or  $p_\pm^\perp > 160$  MeV; at least one photon with  $50^\circ \leq \theta_\gamma \leq 130^\circ$  and  $E_\gamma > 20$  MeV;  $0.1 \leq M_{\pi\pi}^2 \leq 0.85$  GeV<sup>2</sup>. Angles are measured from the electron beam in both codes. Histograms: the 18 KLOE-I observables with 600 uniform bins. The forward–backward asymmetry is  $A = [\sigma(\theta < 90^\circ) - \sigma(\theta > 90^\circ)]/\sigma$ .

### Codes.

- **CEEX-main** (commit e05d007): `pipig -scenario=KLOE-LA -NLO`, one-loop amplitudes from GoSam (`eepipig_noVP_1L_onshell`, `eepipigg`), YFS soft subtraction with cutoff  $E_{\text{min}} = 10^{-4}$  or  $10^{-5}$  GeV; 5 MPI ranks per task.
- **PHOKHARA 10.0** (`PHOKARA_10.0 @ 133d68b`) with the KLOE-I cut file, card `nlo=nlo2=1`, `IFSNLO=1`, `ISR+INT+FSR`, `ivac=0`, `FF_pion=-1` ( $F_\pi = 1$ ), no GVMD, no  $f_0$ , soft cutoff  $w = 10^{-5}$  (soft region  $E_\gamma < w\sqrt{s}$ ). The production binary `phokhara_nostop` only disables the abort on “Max(i) too small” (Section 9); only weighted events are used.
- **BabaYaga** and the earlier PHOKHARA reference set (5 Aug) are taken from the Strong2020 comparison files.

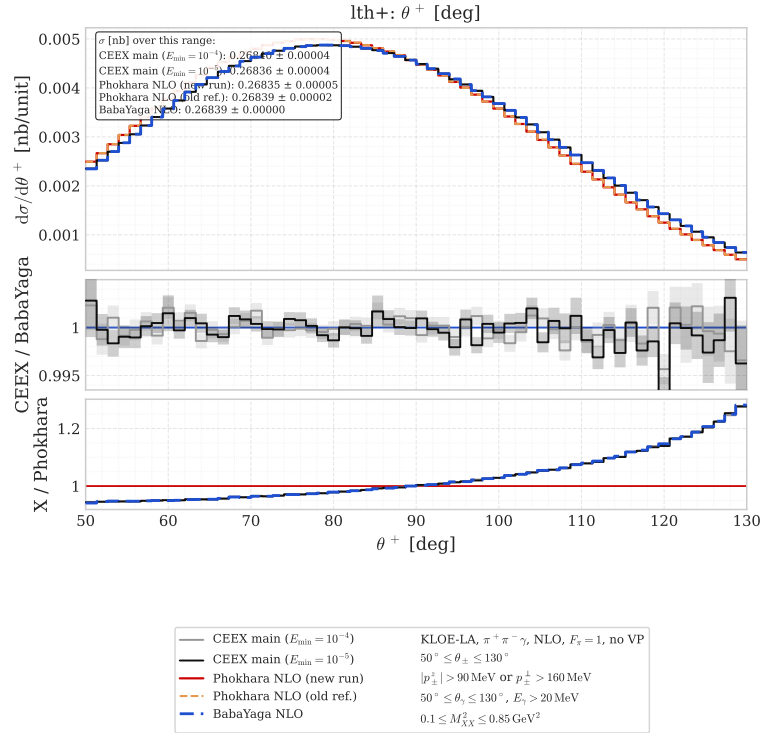
**Pipeline.** SLURM arrays per run name; CEEX rank outputs are screened with `check_total.sh` and outliers are moved aside (`quarantine.sh`) before merging; `plot_compare.py` produces the distribution and ratio plots. Everything is in the repository (Appendix A).

### 3 The discrepancy

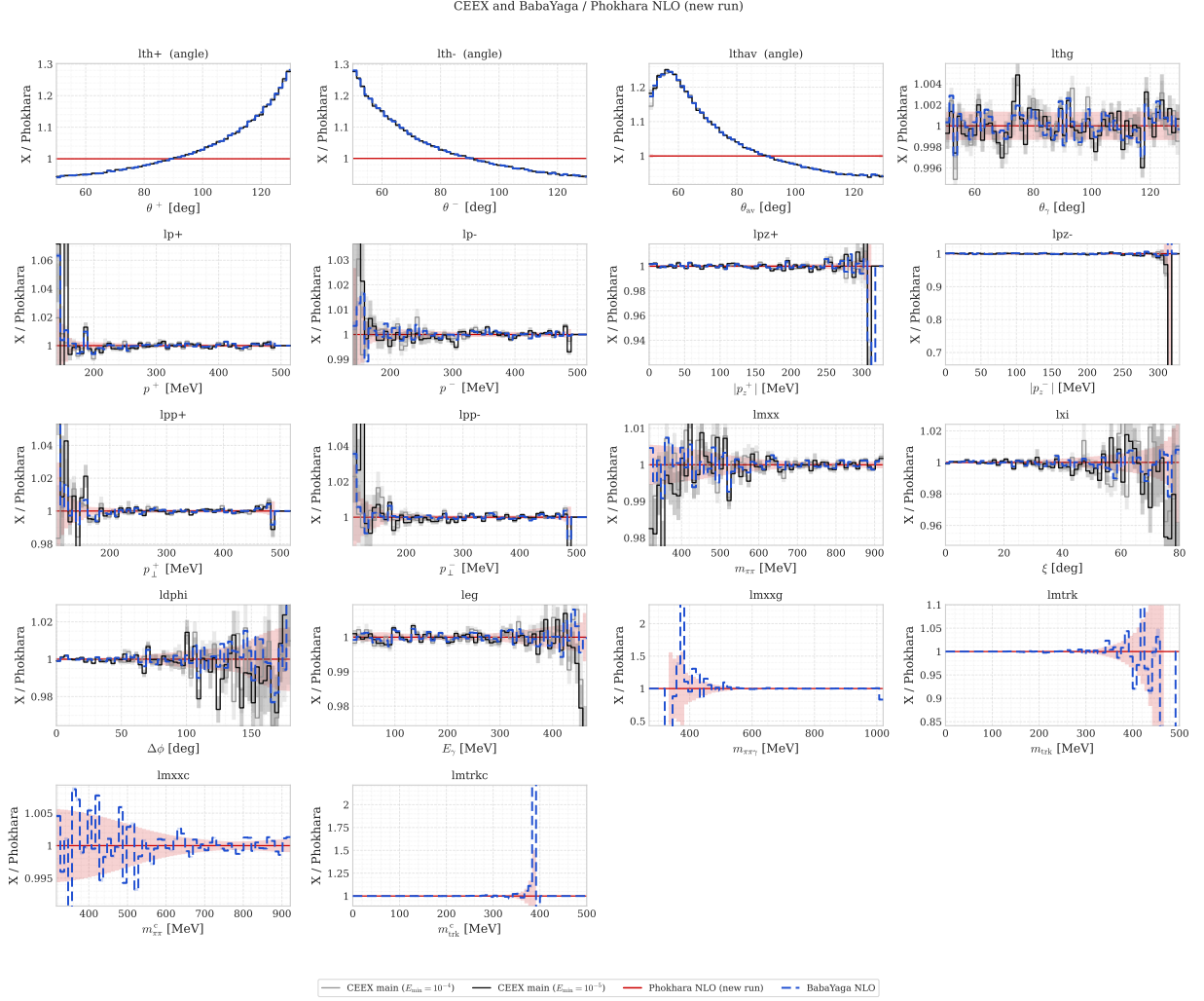
**Table 1:** Production runs (prod500k): PHOKHARA 300 seeds  $\times$  500k events; CEEX 59 tasks  $\times$  5 ranks  $\times$  500k per  $E_{\min}$  (one slow task cancelled, one outlier rank excluded).

|                                | $\sigma$ [nb] | $A(\theta^+)$ |
|--------------------------------|---------------|---------------|
| CEEX, $E_{\min} = 10^{-4}$ GeV | 0.26840(4)    | 0.23751(13)   |
| CEEX, $E_{\min} = 10^{-5}$ GeV | 0.26836(4)    | 0.23752(14)   |
| BabaYaga                       | 0.268389(4)   | 0.23724(1)    |
| PHOKHARA, this work            | 0.268353(46)  | 0.27872(16)   |
| PHOKHARA, earlier reference    | 0.268393(23)  | 0.27861(8)    |

The new PHOKHARA run reproduces the earlier reference. All generators pass the charge-conjugation mirror test  $\theta^+(\theta) = \theta^-(180^\circ - \theta)$ , so this is not a mislabelled charge. Figure 1 shows the  $\theta^+$  distribution and Figure 2 the ratio of every observable to PHOKHARA: only the three pion angles are off, by a smooth tilt from 0.95 to 1.28.



**Figure 1:** Before the fix (prod500k):  $d\sigma/d\theta^+$ , CEEX/BabaYaga, and CEEX and BabaYaga over PHOKHARA (60 bins). CEEX (black, grey) and BabaYaga (blue) agree; PHOKHARA (red) is tilted.



**Figure 2: Before the fix:** every observable, CEEX and BabaYaga over PHOKHARA. Only  $\theta^+$ ,  $\theta^-$  and  $\theta_{\text{avg}}$  are off; the spikes in  $M_{\pi\pi\gamma}$  and  $M_{\text{trk}}$  are empty tail bins.

At LO the codes agree exactly (earlier runs with the Strong2020 form factor:  $A_{\text{LO}} = 0.0779$  and  $\sigma_{\text{LO}} = 2.6523$  nb in both), so the difference lies in the NLO correction to the charge-odd part of the cross section.

## 4 Strategy

The difference is charge-odd ( $C$ -odd): it flips sign under  $\pi^+ \leftrightarrow \pi^-$  and integrates to zero over the symmetric cuts. At  $\mathcal{O}(\alpha^4)$  the  $C$ -odd part of the cross section consists of the gauge-invariant classes  $Q_e^5 Q_\pi^3$  and  $Q_e^3 Q_\pi^5$  (the tree interference is  $Q_e^3 Q_\pi^3$ ). Three tools were used to isolate it:

- **Cutoff independence.** The split into one-photon (Born + virtual + soft photons below the cutoff) and two-photon (hard) parts depends on the cutoff; the sum must not.
- **Matched split.** For the same cutoff each part is uniquely defined, so the two codes must agree part by part. PHOKHARA  $w = 9.804 \times 10^{-6}$  ( $w\sqrt{s} = 10^{-5}$  GeV) matches CEEX  $E_{\min} = 10^{-5}$  GeV.
- **Momentum swap.** At a phase-space point, the  $C$ -odd part of any piece  $X$  is  $X_{\text{odd}} = \frac{1}{2}[X(p_+, p_-) - X(p_-, p_+)]$ . No charge parameter is needed.

Each histogram-level test used enough statistics to resolve  $A(\theta^+)$  to  $\sim 10^{-3}$  while checking that  $\sigma$  and  $M_{\pi\pi}$  are unchanged. The plan with all tests and hypotheses is in `plan.txt`.

## 5 Tests

### 5.1 T1: soft-cutoff scan

**Table 2:** Unpatched PHOKHARA, 30 seeds  $\times$  100k per value of  $w$ .

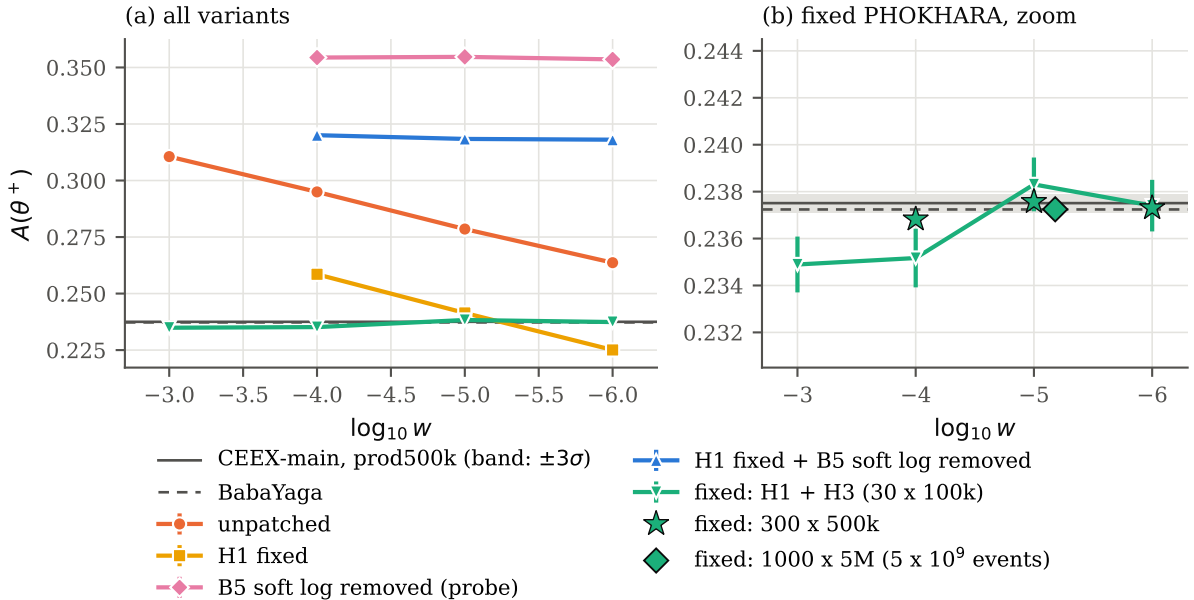
| $w$                     | $10^{-3}$   | $10^{-4}$   | $10^{-5}$   | $10^{-6}$   |
|-------------------------|-------------|-------------|-------------|-------------|
| $\sigma_{1\gamma}$ [nb] | 0.1386      | 0.0955      | 0.0530      | 0.0103      |
| $\sigma_{2\gamma}$ [nb] | 0.1297      | 0.1728      | 0.2149      | 0.2580      |
| $\sigma$ [nb]           | 0.26833(33) | 0.26827(35) | 0.26792(33) | 0.26830(30) |
| $A(\theta^+)$           | 0.3106(11)  | 0.2950(12)  | 0.2785(12)  | 0.2637(11)  |

$A(\theta^+)$  falls linearly in  $\ln w$ , by 0.0156 per decade, while  $\sigma$  is constant to 0.1%: the charge-even soft logarithms cancel between the one- and two-photon parts, a charge-odd one does not. In CEEX,  $E_{\min} = 10^{-4}$  and  $10^{-5}$  GeV give the same  $A(\theta^+)$  to  $10^{-4}$ .

### 5.2 T2: diagnostic variants

Three diagnostic PHOKHARA builds were run at  $w = 10^{-4}, 10^{-5}, 10^{-6}$  (Figure 3a):

- **H1 fixed** (the parenthesis of Section 6.1):  $A(\theta^+)$  moves by  $-0.037$  and keeps its slope. H1 is a  $w$ -independent error.
- **B5 soft log removed** (probe):  $A(\theta^+)$  becomes flat in  $w$ , but at 0.355. The whole  $w$ -dependence comes from this term; removing only the logarithm is not the right correction.
- **Both:** flat, at 0.319.



**Figure 3:**  $A(\theta^+)$  against the PHOKHARA soft cutoff for the unpatched code, the T2 diagnostic variants and the fixed code (H1 + H3), with CEEX (grey band:  $\pm 3\sigma$ ) and BabaYaga. The zoom (b) shows the fixed code at  $30 \times 100k$ , at  $300 \times 500k$  (stars) and at  $1000 \times 5M$  (diamond, drawn just left of  $w = 10^{-5}$ ).

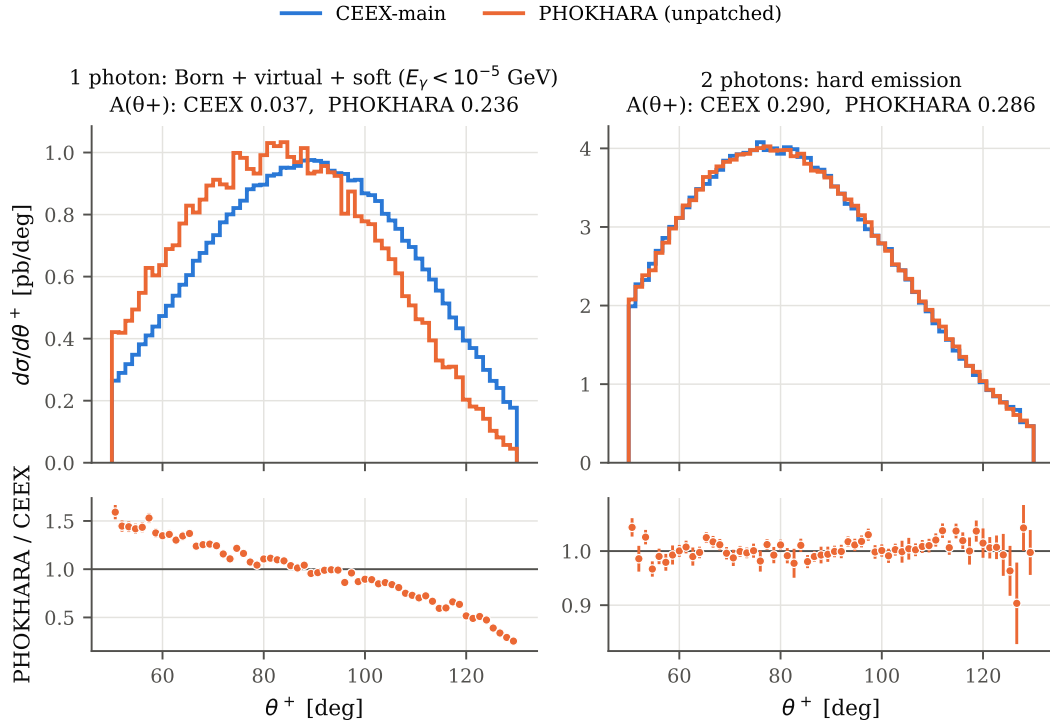
### 5.3 T3: split by photon multiplicity

CEEX-main was given an opt-in output of the per-multiplicity histograms (CEEX\_NPH\_HISTOS=1); PHOKHARA writes them with `print_summed=1`.

**Table 3:** Matched cutoff ( $10^{-5}$  GeV). CEEX 10 seeds  $\times$  5 ranks  $\times$  100k; PHOKHARA 30  $\times$  100k (unpatched).

|                     | $\sigma$ [nb] |             | $A(\theta^+)$ |            | angle shapes<br>$\chi^2/\text{ndf}$ |
|---------------------|---------------|-------------|---------------|------------|-------------------------------------|
|                     | CEEX          | PHOKHARA    | CEEX          | PHOKHARA   |                                     |
| 1 photon (Born+V+S) | 0.05268(7)    | 0.05239(24) | 0.0375(13)    | 0.2360(44) | 79–94                               |
| 2 photons (hard)    | 0.21560(37)   | 0.21588(23) | 0.2901(16)    | 0.2855(10) | 1.0–1.4                             |

The one-photon parts have the same  $\sigma$  and the same  $M_{\pi\pi}$ ,  $\theta_\gamma$  and  $E_\gamma$  shapes, but charge-odd cross sections of 0.0020 (CEEX) and 0.0124 nb (PHOKHARA). The difference, 0.0104 nb, is the whole total discrepancy (Figure 4). The hard two-photon parts agree.

**Figure 4:** T3:  $d\sigma/d\theta^+$  of the one-photon (left) and two-photon (right) contributions at a matched cutoff, with the ratio PHOKHARA/CEEX.

#### 5.4 T4: point-by-point comparison

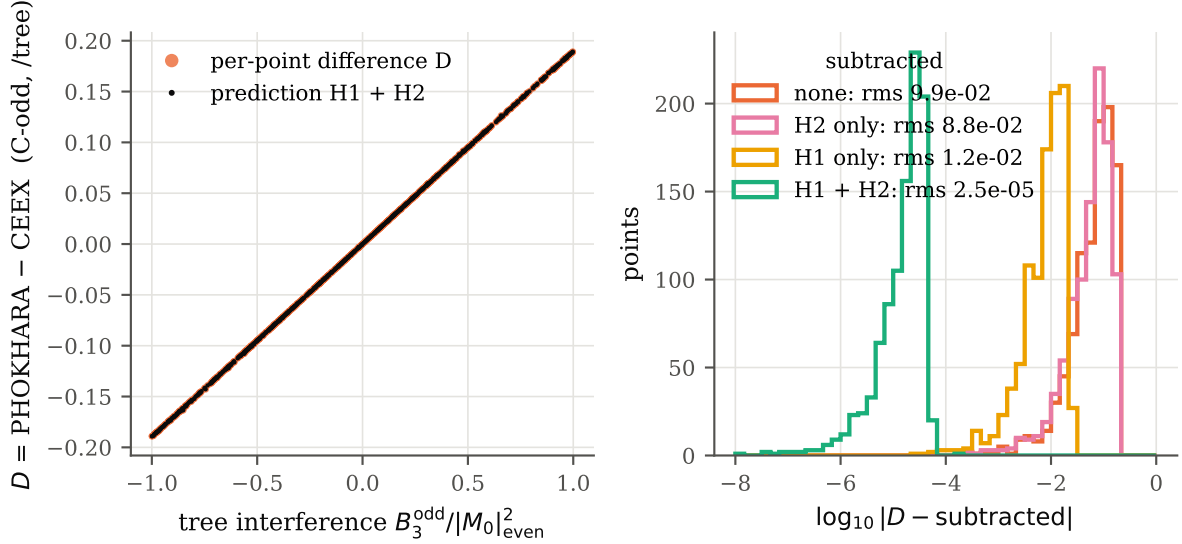
988 accepted one-photon KLOE-LA points were dumped by a PHOKHARA variant (`phokhara_dump`) together with every piece of the one-photon matrix element, at the event and with the pion momenta swapped. A CEEX-main driver (`t4_ceex`) evaluates the tree, the hard virtual  $\frac{\alpha}{4\pi}D$  (GoSam finite part plus mass counterterm), the wave-function counterterm and the YFS factor at the same momenta; the pion mass is set to PHOKHARA’s value. Each code is normalised to its own charge-even tree:

- tree  $C$ -odd/ $C$ -even ratio: identical to  $7 \times 10^{-13}$ ;
- $C$ -even NLO correction: CEEX  $-0.77566$ , PHOKHARA  $-0.77570$ , per-point difference rms  $10^{-5}$ ;
- $C$ -odd NLO correction: per-point difference  $D$  with rms 0.099;
- $C$ -odd soft (eikonal/YFS) part: agrees, rms  $7 \times 10^{-4}$ .

$D$  is exactly proportional to the tree interference at every point. Subtracting the two errors, each computed exactly per point ( $v_s = 0.29655$ ,  $v_f = 0.023\text{--}0.036$ ,  $S_{\text{ISR}}(w) = -0.95556$ ),

$$\text{pred}_{\text{H1}} = B_4 - \frac{1}{2}B_3(v_f + v_s), \quad \text{pred}_{\text{H2}} = \frac{1}{2}B_3(1 + S_{\text{ISR}}(w)),$$

leaves a residual of rms  $2.5 \times 10^{-5}$  with mean  $10^{-6}$  (Figure 5). H1 alone leaves  $1.2 \times 10^{-2}$ , H2 alone  $8.8 \times 10^{-2}$ .



**Figure 5:** T4, 988 points. Left: the  $C$ -odd difference  $D$  against the tree interference, with the H1+H2 prediction. Right:  $\log_{10} |D - \text{subtracted}|$  for no subtraction, H1, H2 and H1+H2.

## 6 The two errors in PHOKHARA

The one-photon weight in the full-NLO mode is `amplit+helicityampL0+NLOpions+virt_pions`. In `ampL0` (inside `helicityampL0`) the pion branch adds, per photon polarisation:  $B1 = |F|^2(1 + v_s)$ ,  $B2 = |I|^2v_f$ ,  $B3 = 2\text{Re}[I^*F]$  (tree interference),  $B4$  and  $B5$  below. Here  $I$  and  $F$  are the ISR and FSR tree amplitudes,  $v_s$  the electron-vertex correction at  $s$  and  $v_f$  the pion-vertex correction at  $q^2$ , both  $|M|^2$ -level factors with the analytic soft part removed.

### 6.1 H1: misplaced factor in the $\text{ISR} \times \text{FSR}$ correction

```

if((nlo2.eq.1).and.(pion.eq.1))then
  ampL0 = ampL0
  4    + dreal(
  5      dconjg(dme*(mb(1,1)-ma(1,1))*cvac_qq)*dme*(ddpl(1,1)-ddmi(1,1))*cvac_s    ! T1
  6      + dconjg(dme*(mb(2,2)-ma(2,2))*cvac_qq)*dme*(ddpl(2,2)-ddmi(2,2))*cvac_s    ! T2
  7      + dconjg((-ebppb*ma(2,1)+el_m2/ebppb*mb(2,1))*cvac_qq)
  8      *(-ebppb*ddmi(2,1)+el_m2/ebppb*ddpl(2,1))*cvac_s                        ! T3
  9      + dconjg((ebppb*mb(1,2)-el_m2/ebppb*ma(1,2))*cvac_qq)
  1     *(ebppb*ddpl(1,2)-el_m2/ebppb*ddmi(1,2))*cvac_s                        ! T4
  2     *(ver_f+ver_s) )                ! multiplies T4 only

```

(Continuation lines re-indented for reading; the code is unchanged.) The block is meant to be  $\text{Re}[T_1 + T_2 + T_3 + T_4](v_f + v_s)$ , the amplitude-level corrections on each side of  $2\text{Re}[I^*F]$ . As written it is  $\text{Re}[T_1 + T_2 + T_3] + \text{Re}[T_4](v_f + v_s)$ , so  $\text{Re}[T_1 + T_2 + T_3]$  (about a third of the tree interference) is added with coefficient 1. The muon version of the same block (`phokhara_10.0.f:10384-10394`) is correct: `2*dreal(sum)*(ver_s+ver_f)/2`.

**Fix** (`variants/h1.patch`): `*cvac_s *(ver_f+ver_s) )`  $\rightarrow$  `*cvac_s *(ver_f+ver_s).`

## 6.2 H2: tree and analytic ISR soft function in the ISR one-loop block

Block B5 contracts the FSR tree with the ISR current at one loop:

```
ampL0=ampL0+dreal( dconj(... ddp1, ddmi ...)*( ... apl, ami ... )    ! B5
```

The current (`hard_soft_part/ISRcurrent_no_mass_terms.f`) has the coefficient

```
V1(1)=2.d0*log(4.d0*w**2)*(log(m2)+1.d0) -4.d0*log(qq2)+3.d0*log(m2) + ...    ! 1.134
V1(1)=(1.d0-2.d0*alpha/4.d0/pi*V1(1))/Sp                                ! 1.210
```

It therefore contains the tree (the “1”) and the analytic ISR soft function, and B5 carries an extra  $\text{Re}[F^*I](1 + S_{\text{ISR}}(w))$  with

$$S_{\text{ISR}}(w) = \frac{\alpha}{\pi} \left[ -\ln(4w^2) (1 + L) - \frac{1}{2}L^2 - L - \frac{\pi^2}{3} \right], \quad L = \ln \frac{m_e^2}{s}.$$

- The tree interference is already added by the `fsr=2` block (B3).
- In the full-NLO mode the analytic ISR soft function is replaced by the eikonal in `NLOptions`; it is subtracted from  $v_s$ ,  $v_f$  and the leptonic tensor, but not here.
- At  $w \simeq 10^{-5}$  the two parts nearly cancel ( $1 + S_{\text{ISR}} = 0.044$ ), but  $S_{\text{ISR}}$  changes by 0.152 per decade of  $w$ . This gives the T1 slope:  $\frac{1}{2} \cdot 0.152 \cdot A_{\text{LO}} \approx 0.019$  per decade, against 0.0156 measured.

**Fix (variants/h3.patch):** subtract  $\text{Re}[\sum T](1 + S_{\text{ISR}})$  after B5, with  $S_{\text{ISR}}$  computed as in the  $v_s$  subtraction.

## 7 Result with the fix

`phokhara_fix` is the production binary with `h1.patch` and `h3.patch`.

**Table 4:** Fixed PHOKHARA against the soft cutoff.

| $w$                                                                          | $10^{-3}$   | $10^{-4}$    | $10^{-5}$    | $10^{-6}$    |
|------------------------------------------------------------------------------|-------------|--------------|--------------|--------------|
| <i>30 seeds <math>\times</math> 100k</i>                                     |             |              |              |              |
| $\sigma$ [nb]                                                                | 0.26799(33) | 0.26892(35)  | 0.26854(32)  | 0.26784(31)  |
| $A(\theta^+)$                                                                | 0.2349(12)  | 0.2352(13)   | 0.2383(12)   | 0.2374(11)   |
| <i>300 seeds <math>\times</math> 500k</i>                                    |             |              |              |              |
| $\sigma$ [nb]                                                                | –           | 0.268345(48) | 0.268351(46) | 0.268354(44) |
| $A(\theta^+)$                                                                | –           | 0.23682(17)  | 0.23757(17)  | 0.23730(16)  |
| <i>1000 seeds <math>\times</math> 5M (<math>5 \times 10^9</math> events)</i> |             |              |              |              |
| $\sigma$ [nb]                                                                | –           | –            | 0.268398(8)  | –            |
| $A(\theta^+)$                                                                | –           | –            | 0.23725(3)   | –            |

At high statistics the slope in  $w$  is  $-0.00023 \pm 0.00012$  per decade (unpatched:  $+0.0156$ ), and the  $w$ -averaged asymmetries are

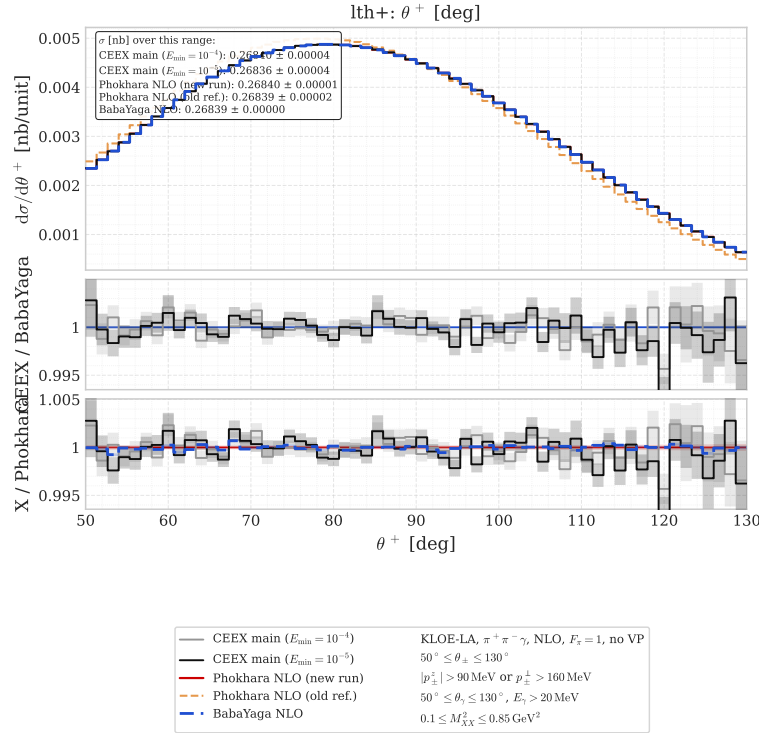
|                                                    | $A(\theta^+)$ | $-A(\theta^-)$ | $-A(\theta_{\text{avg}})$ |
|----------------------------------------------------|---------------|----------------|---------------------------|
| PHOKHARA, fixed (300 $\times$ 500k, $w$ -average)  | 0.23724(10)   | 0.23722(10)    | 0.26651(9)                |
| PHOKHARA, fixed (1000 $\times$ 5M, $w = 10^{-5}$ ) | 0.23725(3)    | 0.23728(3)     | 0.26651(3)                |
| CEEX, $E_{\text{min}} = 10^{-4}$ GeV               | 0.23751(13)   | 0.23723(13)    | 0.26660(13)               |
| BabaYaga                                           | 0.23724(1)    | 0.23727(1)     | 0.26650(1)                |

At  $w = 10^{-5}$  and  $10^{-6}$  the fixed PHOKHARA agrees with CEEX and BabaYaga within  $0.3$ – $1.5\sigma$ . The  $w = 10^{-4}$  sample lies  $0.0007$  lower. To check it, the multiplicity split of Section 5.3 was repeated with the fixed code at a matched cutoff of  $10^{-4}$  GeV (PHOKHARA  $w = 9.804 \times 10^{-5}$ , 100 seeds  $\times$  500k; CEEX  $E_{\text{min}} = 10^{-4}$  GeV):

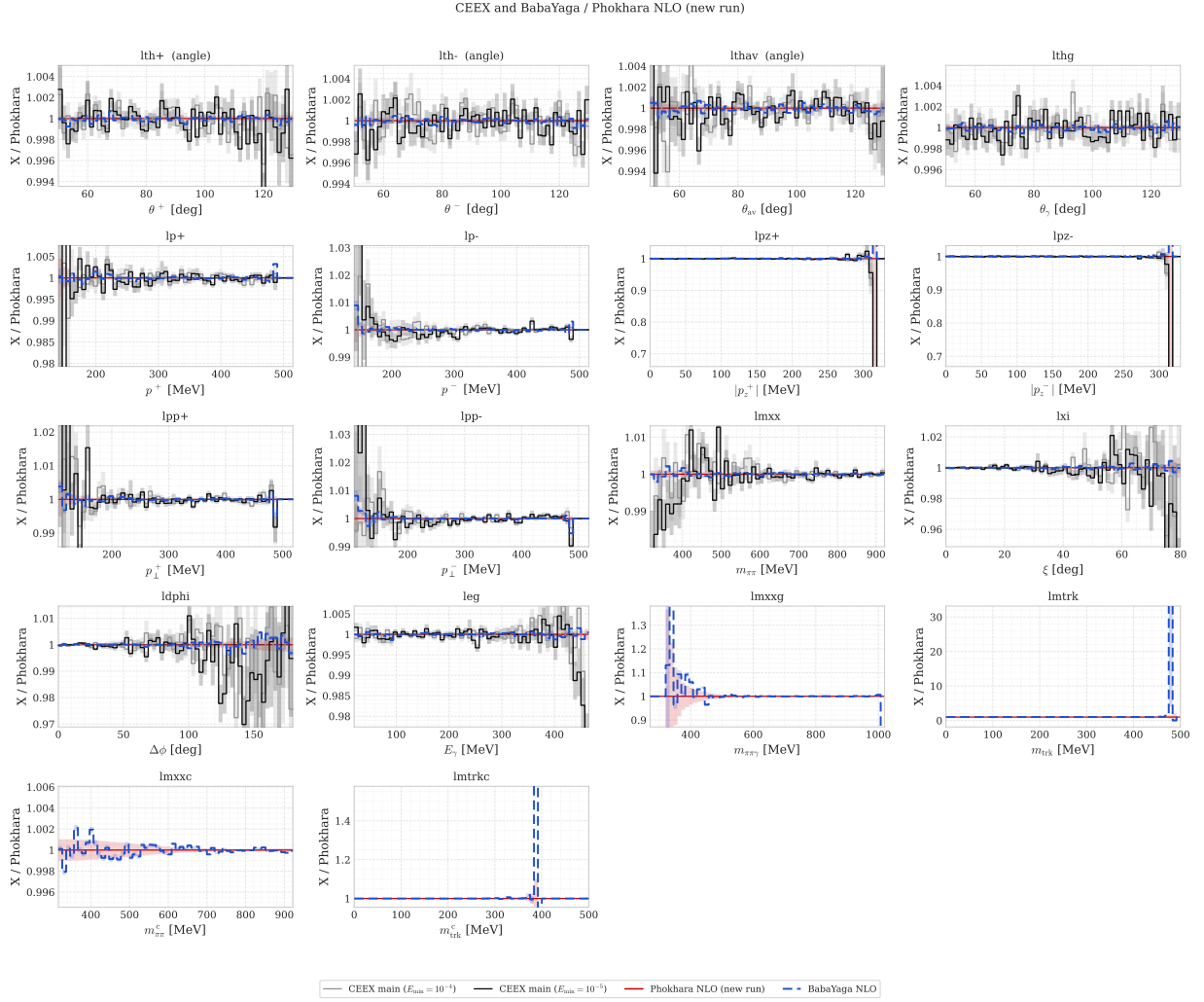
|                     | CEEX        | $\sigma$ [nb]<br>PHOKHARA (fixed) | CEEX       | $A(\theta^+)$<br>PHOKHARA (fixed) |
|---------------------|-------------|-----------------------------------|------------|-----------------------------------|
| 1 photon (Born+V+S) | 0.09533(10) | 0.09539(7)                        | 0.1548(11) | 0.1558(7)                         |
| 2 photons (hard)    | 0.17290(23) | 0.17301(5)                        | 0.2826(13) | 0.2829(2)                         |
| sum                 | 0.26823(25) | 0.26840(8)                        | 0.2372(9)  | 0.2377(3)                         |

Both parts agree, with angle-shape  $\chi^2/\text{ndf} = 0.7\text{--}1.4$ , and this sample gives  $A(\theta^+) = 0.2377(3)$ , consistent with  $w = 10^{-5}$  and  $10^{-6}$ . A 2% change of the cutoff cannot move  $A(\theta^+)$  by 0.0009, so the low  $w = 10^{-4}$  value is a fluctuation of that sample ( $2.6\sigma$  from this one). T4 at  $w = 10^{-4}$  also leaves no systematic  $C$ -odd residual after H1+H2 (mean  $-7 \times 10^{-6}$ ).

**Highest statistics.** A final run of  $1000 \times 5\text{M}$  events ( $5 \times 10^9$  events passing the cuts,  $w = 10^{-5}$ ) gives  $\sigma = 0.268398(8)$  nb and  $A(\theta^+) = 0.23725(3)$ ,  $-A(\theta^-) = 0.23728(3)$ ,  $-A(\theta_{\text{avg}}) = 0.26651(3)$ . Its statistical error is four times below that of CEEX (prod500k,  $\pm 0.00013$ ), so the comparison is now limited by CEEX. Figures 6 and 7 repeat Figures 1 and 2 with this run: the pion-angle shape  $\chi^2/\text{ndf}$  against CEEX drops from  $\sim 1900$  to 1.0–1.6, and against BabaYaga it is 0.9–1.2.



**Figure 6: After the fix:** as Figure 1, with the fixed PHOKHARA ( $1000 \times 5\text{M}$ ,  $w = 10^{-5}$ ) as the denominator. The tilt is gone.



**Figure 7:** After the fix: every observable, CEEX and BabaYaga over the fixed PHOKHARA (1000×5M). With  $5 \times 10^9$  PHOKHARA events the band around 1 is its own error; the scatter is that of CEEX.

## 8 Caveats and open points

- Only the configuration studied here was tested ( $F_\pi = 1$ , no VP, KLOE-LA cuts). Both errors are on code paths active for `pion=1` with `nlo2=1` in any configuration, so they should also affect the Strong2020 form-factor results (there the NLO gap in  $A(\theta^+)$  is 0.012: CEEX 0.0804, PHOKHARA 0.0924). This has not been re-run.
- After the fix, one of the three high-statistics samples ( $w = 10^{-4}$ ) lies 0.0007 below the others; an independent sample at the same cutoff agrees with them and with CEEX part by part (Section 7), so it is treated as a fluctuation.
- Muons take a different, correct branch and are not affected.
- Totals and charge-even distributions are not affected by either error ( $\sigma$  unchanged in every run; charge-even per-point agreement  $10^{-5}$ ).
- CEEX lies 0.00027 ( $2\sigma$  at each  $E_{\min}$ ) above BabaYaga in  $A(\theta^+)$ , and the fixed PHOKHARA at  $5 \times 10^9$  events sides with BabaYaga: 0.23725(3) against 0.23724(1), with  $-A(\theta^-)$  and  $-A(\theta_{\text{avg}})$  equal to  $10^{-5}$  as well. A possible 0.1% CEEX effect in  $A(\theta^+)$  would now need more CEEX statistics to confirm; it is 150 times smaller than the original PHOKHARA effect.  $-A(\theta^-)$  and  $-A(\theta_{\text{avg}})$  of CEEX agree with both.

- With  $5 \times 10^9$  PHOKHARA events, small BabaYaga–PHOKHARA shape differences become visible in a few charge-even observables ( $\chi^2/\text{ndf}$  2–3 in  $p_\pm$ ,  $p_\pm^\perp$ ,  $E_\gamma$ ; 10 in  $|p_-^z|$ , where BabaYaga’s binning already gave  $\sim 20$  before). CEEX agrees with PHOKHARA there at  $\chi^2/\text{ndf} \approx 1$ . They are unrelated to the pion-angle issue.
- Tests T5 (charge-class split) and T6 (two-photon point-by-point) of the plan were not needed: the two-photon part agrees (T3) and H1+H2 explain the one-photon part (T4).

## 9 Other findings

- The checked-out PHOKARA\_10.0 has an empty stub `variables_cuts.f` (no cuts, no histograms); the KLOE-I cut file comes from the Strong2020 scenario repository. The card format and the `FF_pion` numbering changed with respect to the Strong2020 cards.
- PHOKHARA aborts without writing histograms when a weight exceeds the scanned maximum (`phokhara_10.0.f:1910,2135,2263`). Dropping such seeds, as the Strong2020 merge did, biases the result; for weighted events the abort is unnecessary.
- `virt_pions` overwrites its quad-precision momentum arguments ( $\sim 1$  GeV). This is harmless in production but breaks any second call in the same event.
- In the 5 Aug reference folder, `Phokhara/NL0novp_*.csv` is the muon result (0.9925 nb), not the pion one.
- CEEX: a few rank outputs per hundred diverge at 100k events (11/300); they are removed before merging. The CEEX-main README still documents `-ref_alloc` and `-vfreeze`, which main no longer parses.
- Cost: at equal precision on  $\sigma$  (prod100k), PHOKHARA needed about 45 times less CPU than CEEX: 0.25 against 28 CPU-hours.

## A Reproduction and files

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|                                                              |                                                                                                                                                                                                               |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>plan.txt</code>                                        | test plan (T0–T6, hypotheses H1–H5)                                                                                                                                                                           |
| <code>RUN/config.sh</code>                                   | all paths                                                                                                                                                                                                     |
| <code>RUN/RUN_CEEEX/</code> , <code>RUN/RUN_PHOKHARA/</code> | SLURM submitters and job scripts, runcards                                                                                                                                                                    |
| <code>RUN/RUN_PHOKHARA/variants/</code>                      | <code>h1.patch</code> , <code>h3.patch</code> (the fix), <code>h2.patch</code> (T2 probe), <code>dump.patch</code> (T4), <code>build.sh</code>                                                                |
| <code>RUN/RUN_CEEEX/t4/</code>                               | CEEX point driver; <code>MC_nphsplit.patch</code> (per-multiplicity histograms)                                                                                                                               |
| <code>RUN/POST_PROCESS/</code>                               | <code>merge</code> , <code>outlier check</code> , <code>phok_table.py</code> , <code>t3_compare.py</code> , <code>t4_compare.py</code>                                                                        |
| <code>RESULTS/</code>                                        | <code>prod500k</code> , <code>t1_t2</code> , <code>t3_split</code> , <code>t4_points</code> , <code>t4_fix</code> , <code>fix500k</code> , <code>final500k</code> , <code>fix5M</code> , <code>final5M</code> |
| <code>REPORT/</code>                                         | this report, <code>make_figures.py</code> , <code>make_numbers.py</code>                                                                                                                                      |

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Building the fixed binary:

```
cd RUN/RUN_PHOKHARA/variants
./build.sh fix h1.patch h3.patch          # -> src/phokhara_fix
cd ../../
EXE=RUN_PHOKHARA/src/phokhara_fix \
  RUN_PHOKHARA/submit.sh <run> <n_seeds> <nges> 10000 <walltime>
```